CLOUD NATIVE

```
PS D:\kitchen-app> kubectl get nodes
NAME             STATUS   ROLES           AGE     VERSION
docker-desktop   Ready    control-plane   5d19h   v1.34.1
PS D:\kitchen-app>
```

```
PS D:\kitchen-app> kubectl apply -f k8s/namespace.yaml
namespace/kitchen-app unchanged
PS D:\kitchen-app> kubectl apply -f k8s/db/statefulset.yaml
service/mongo unchanged
statefulset.apps/mongo configured
PS D:\kitchen-app> kubectl apply -f k8s/api/deployment.yaml
service/api unchanged
deployment.apps/api-deployment unchanged
PS D:\kitchen-app> kubectl apply -f k8s/web-ui/deployment.yaml
configmap/web-ui-config unchanged
deployment.apps/web-ui-deployment unchanged
service/web-ui-service unchanged
PS D:\kitchen-app>
```

```
PS D:\kitchen-app> kubectl port-forward -n kitchen-app svc/api 5050:3000
Forwarding from 127.0.0.1:5050 -> 3000
Forwarding from [::1]:5050 -> 3000
```

```
PS D:\kitchen-app> Invoke-RestMethod -Method Get -Uri "http://localhost:5050/health"

status db
------ --
ok     connected

PS D:\kitchen-app>
```

```
PS D:\kitchen-app> Invoke-RestMethod -Method Post -Uri "http://localhost:5050/orders" -C
ontentType "application/json" -Body (@{ dish = "Paneer Tikka" } | ConvertTo-Json)


_id             : 69be8742fe6ef9b141391c02
dish            : Paneer Tikka
status          : pending
createdAt       : 2026-03-21T11:55:46.747Z
statusUpdatedAt : 2026-03-21T11:55:46.747Z
nextStatus      : preparing
nextStatusAt    : 2026-03-21T11:55:51.747Z



PS D:\kitchen-app>
```

```
PS D:\kitchen-app> kubectl port-forward -n kitchen-app svc/web-ui-service 8080:80
Forwarding from 127.0.0.1:8080 -> 80
Forwarding from [::1]:8080 -> 80
```

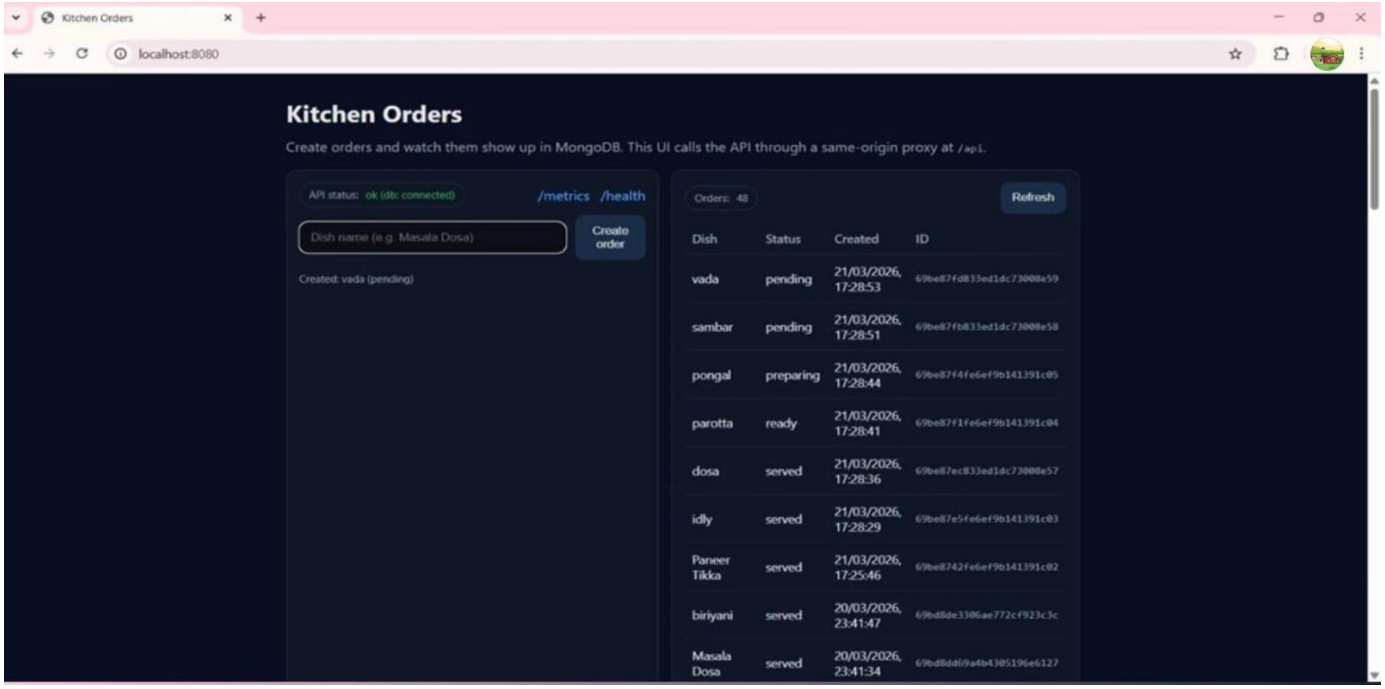
```
PS D:\kitchen-app> kubectl port-forward -n kitchen-app svc/prometheus-service 9090:9090
Forwarding from 127.0.0.1:9090 -> 9090
Forwarding from [::1]:9090 -> 9090
```

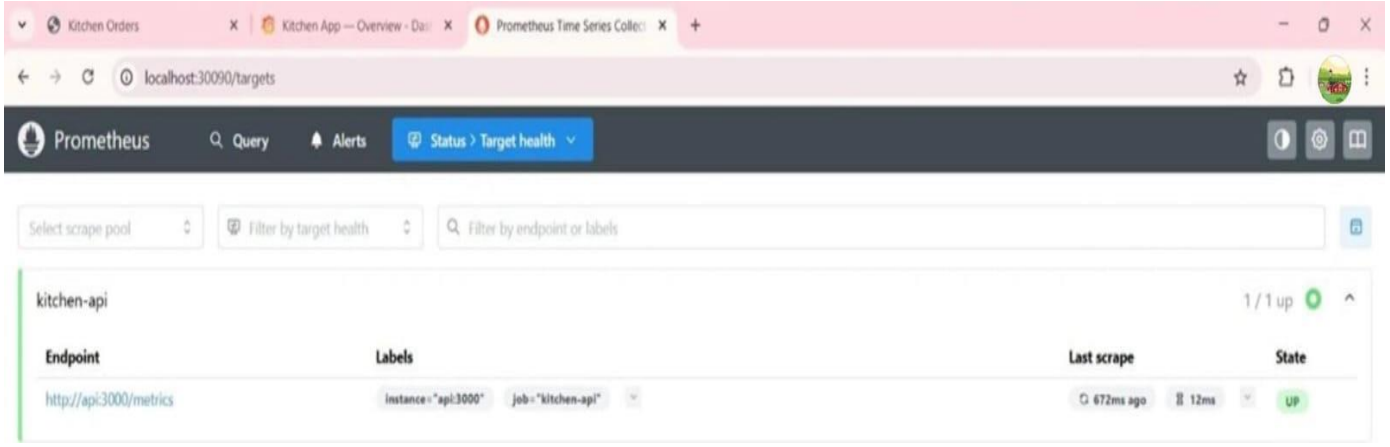
```
PS D:\kitchen-app> kubectl port-forward -n kitchen-app svc/grafana-service 3000:3000
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
```

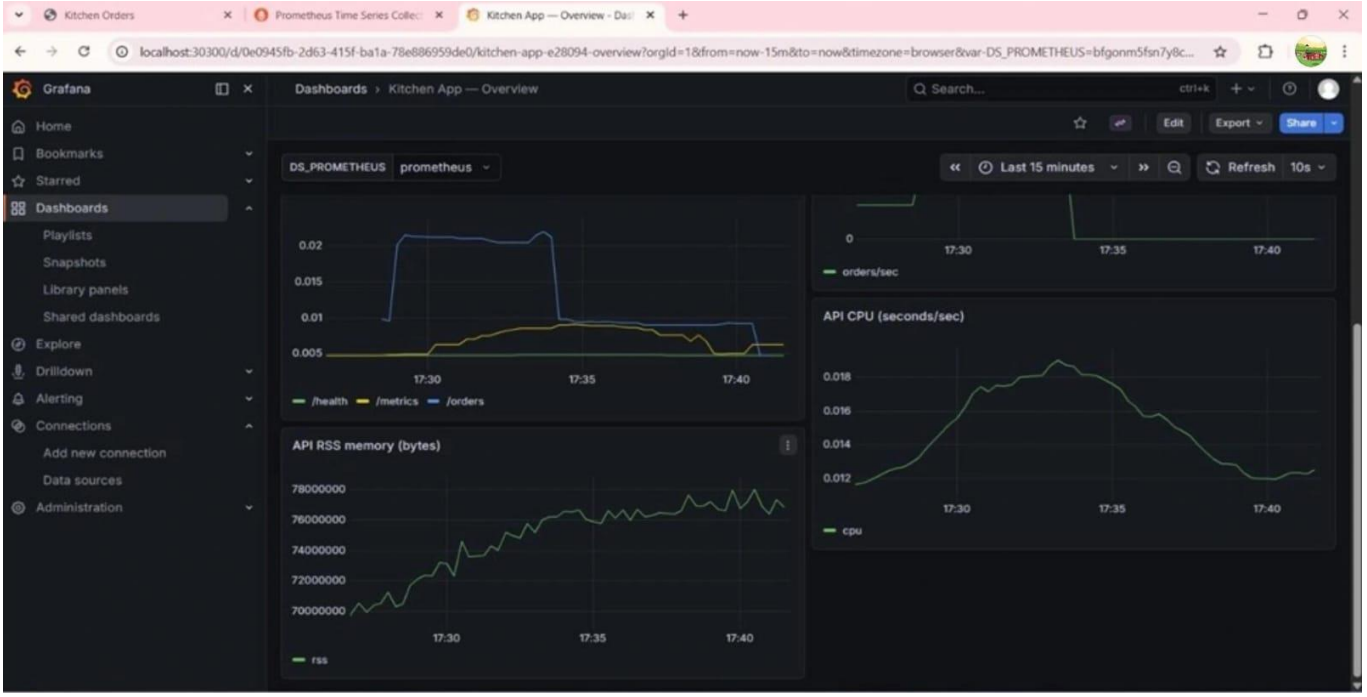
```
PS D:\kitchen-app> kubectl get svc -n kitchen-app
NAME                TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)          AGE
api                 ClusterIP   10.97.157.111    <none>        3000/TCP         5d19h
grafana-service     NodePort    10.102.228.66    <none>        3000:30300/TCP   3d19h
mongo               ClusterIP   10.103.225.209   <none>        27017/TCP        5d19h
prometheus-service  NodePort    10.100.225.116   <none>        9090:30090/TCP   3d19h
web-ui-service      NodePort    10.111.141.62    <none>        80:30080/TCP     5d19h
PS D:\kitchen-app>
```

```
PS D:\kitchen-app> kubectl get pods -n kitchen-app
NAME                                  READY   STATUS    RESTARTS          AGE
api-deployment-b5b478648-p78zz        1/1     Running   1 (17h ago)       18h
api-deployment-b5b478648-vclj8        1/1     Running   1 (17h ago)       18h
grafana-deployment-56d8f6f9b-dzgvw    1/1     Running   1 (17h ago)       18h
mongo-0                               1/1     Running   6 (17h ago)       5d19h
prometheus-deployment-68ffcff5dd-dwvgl 1/1    Running   1 (17h ago)       18h
web-ui-deployment-788bf7774c-967dq    1/1     Running   13 (5m47s ago)    5d19h
web-ui-deployment-788bf7774c-ww5c7    1/1     Running   13 (6m5s ago)     5d19h
PS D:\kitchen-app>
```

OUTPUT


Kitchen Orders
localhost:8080
Kitchen Orders
Create orders and watch them show up in MongoDB. This UI calls the API through a same-origin proxy at /api.
API status: ok (db: connected)
/metrics /health
Dish name (e.g. Masala Dosa)
Create order
Created: vada (pending)
Orders: 48
Refresh
Dish Status Created ID
vada pending 21/03/2026, 17:28:53
sambar pending 21/03/2026, 17:28:51
pongal preparing 21/03/2026, 17:28:44
parotta ready 21/03/2026, 17:28:41
dosa served 21/03/2026, 17:28:36
idly served 21/03/2026, 17:28:29
Paneer Tikka served 21/03/2026, 17:25:46
biriyani served 20/03/2026, 23:41:47
Masala Dosa served 20/03/2026, 23:41:34


Kitchen Orders
Kitchen App — Overview - Dat
Prometheus Time Series Collec
localhost:30090/targets
Prometheus
Query
Alerts
Status > Target health
Select scrape pool
Filter by target health
Filter by endpoint or labels
kitchen-api
1 / 1 up
Endpoint
Labels
Last scrape
State
http://api:3000/metrics
instance="api:3000"
job="kitchen-api"
672ms ago
12ms
UP


Kitchen Orders
Prometheus Time Series Collec
Kitchen App — Overview - Das
localhost:30300/d/0e0945fb-2d63-415f-ba1a-78e886959de0/kitchen-app-e28094-overview?orgId=1&from=now-15m&to=now&timezone=browser&var-DS_PROMETHEUS=bfgonm5fsn7y8c...
Grafana
Dashboards › Kitchen App — Overview
Search...
ctrl+k
Edit
Export
Share
Home
Bookmarks
Starred
Dashboards
Playlists
Snapshots
Library panels
Shared dashboards
Explore
Drilldown
Alerting
Connections
Add new connection
Data sources
Administration
DS_PROMETHEUS
prometheus
Last 15 minutes
Refresh
10s
0.02
0.015
0.01
0.005
17:30
17:35
17:40
/health
/metrics
/orders
0
orders/sec
API CPU (seconds/sec)
0.018
0.016
0.014
0.012
cpu
API RSS memory (bytes)
78000000
76000000
74000000
72000000
70000000
rss